18ME15 / 18ME25 / 16ME15 / 16ME25 / 12ME15 / 12ME25

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I / II Semester B. E. Fast track Examinations Oct-2020

Common to all Branches

ELEMENTS OF MECHNAICAL ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of steam tables is permitted.

PART-A

1	1.1	A transition in which a system changes from one state to another state is called _____.	01
	1.2	The thermodynamic properties which depend on the mass of the system are called _____ properties.	01
	1.3	Determine the quality of steam for $p = 10\ bar$ and $v = 0.180\ m^3/kg$.	02
	1.4	Classify the water turbine based on the type of energy at inlet.	01
	1.5	Kaplan turbine is an _____ flow reaction turbine having small number of blades.	01
	1.6	Name any four commonly used refrigerants.	02
	1.7	During power stroke of four stroke petrol engine, the piston moves from _____ to _____.	01
	1.8	Justify, why 4-stroke diesel engines are called as constant pressure cycle engine?	01
	1.9	Define stroke and clearance volume of an IC engine.	02
	1.10	Cylindrical holes are drilled by means of a revolving tool called _____.	01
	1.11	In _____ milling operation, two parallel surfaces of work piece can be milled simultaneously.	01
	1.12	What are the functions of tailstock of lathe machine?	02
	1.13	In oxyacetylene welding, _____ is used to clean the surface from contaminating oxides.	01
	1.14	_____ belt drive is used to connect shafts which are parallel and rotating in opposite directions.	01
	1.15	List the advantages of producing finished parts using CNC machine.	02

PART-B

2	a	What are the similarities and dis-similarities between the work and heat?	04
	b	Explain and represent the following on temperature –enthalpy diagram of formation of steam. i) Sensible heat (h_f) ii) Latent heat (h_{fg}) iii) Enthalpy of superheated steam.	06
	c	Steam at a pressure of $8\,bar$ has a temperature 200℃. What is its specific enthalpy and specific volume? Assume the specific heat of steam to be $2.25\,kJ\,/kg\,K$.	06
3	a	Draft tube is provided in which type of turbines? Give reasons for the same.	04
	b	Sketch and explain the working principle of Pelton wheel.	06
	c	What are refrigerants? Mention the following properties that an ideal refrigerator should possess. i) Thermodynamic properties ii) Physical properties.	06
		OR	
4	a	Distinguish between the following turbines. i) Impulse and Reaction water turbine ii) Francis turbine and Kaplan turbine.	08
	b	Name the various parts of a vapour refrigerator compression and state their functions.	08
5	a	Differentiate between two stroke and four stroke $I.C$ engine with respect to the following: i) Completion of cycle ii) Power stroke. iii) Fuel consumption iv) Mechanical efficiency.	08
	b	A 4-cylinder two stroke petrol engine develops $26kW$ brake power at $2200\,rpm$. The mean effective pressure is 7 bar and mechanical efficiency is 87%. Determine the bore diameter , the stroke of the engine and the torque if stroke length is 1.5 times the bore and the speed of the engine is $450rpm$.	08
		OR	
6	a	With the help of a PV diagram, explain the working of four stroke petrol engine.	08
	b	The following are the details of 4-stroke petrol engine: Brake diameter of the brake drum= $600.3mm$, Full brake laod on drum= $250N$, brake drum speed= $450rpm$, calorific value of petrol= $40\,MJ/kg$, Brake thermal efficiency=32%, Mechanical efficiency= 80%, specific gravity of petrol= 0.82. Determine: i) Brake Power ii) Indicated Power iii) Fuel consumption in litres per second iv) Indicated thermal efficiency.	08

7	a	What is lathe? Enlist the classification of lathe according to their construction and design.	04
	b	With a neat sketch, explain the following operations. i) Plain and Taper turning on Lathe machine ii) Counter boring and Counter sinking on Drilling machine.	06
	c	Explain the following with a neat sketch: i) Up-milling ii) Down-milling.	06
8	a	What are the functions of coated electrodes of arc welding?	04
	b	With a neat sketch explain different types of flames, which is obtained by regulating the ratio of oxygen to acetylene.	06
	c	Write a note on the following phenomenon with respect to power transmission: i) Creep ii) Slip iii) Idler Puley.	06